



Assignment # 4

Subject: Database Systems -CS2005

Post Date: 20/11/2025

Total Marks: 50

Due Date: **7/12/2025**

Course Instructors: Dr. Zulfiqar, Dr. Farrukh Salim, Basit Jasani, Ali Naseer, Atiya Jokhio, Javeria Farooq, Hajira Ahmed, Huzaifa, Umair

Instructions to be strictly followed.

- For all questions involving SQL Queries:
 - o **Submit the SQL Scripts in a .txt file.**
- It should be obvious that submitting your work after the due date will result in zero points being awarded.
- Plagiarism (copying/cheating) and late submissions result in a zero mark.

Question #1:

Consider the following scenario:

A ride-hailing app (like Careem/Uber) processes a transaction where a customer books a ride, the driver accepts it, and payment is reserved from the wallet.

During this process:

- The driver accepted the ride (system recorded).
- The customer's wallet balance was deducted.
- But before the ride confirmation was shown to the customer, the server crashed.

Tasks:

- Identify which ACID property/properties were violated.
- Explain *why* the violation happened using technical reasoning.
- Rewrite the same scenario in a way that *preserves* all ACID properties.
- What could be the long-term business consequences if such violations occur frequently?

Question #2:

Two transactions run concurrently on an online inventory system:

T1: Update quantity of product P1 after a sale

Read(P1.quantity)

Write(P1.quantity – sold_units)

T2: Update quantity of product P1 after a return

Read(P1.quantity)

Write(P1.quantity + returned_units)

Because both are running together, the final quantity becomes inconsistent.

Tasks:

- Write a **non-serial schedule** representing this situation.
- Construct the **precedence graph** step-by-step.
- Determine whether the schedule is conflict-serializable or not.
- If the schedule is not serializable, rewrite a corrected schedule that ensures consistency.
- What real-world loss might the company face due to such anomalies?

Question #3:

Consider the four transactions T1, T2, T3 and T4 and the schedules S1 and S2 given below.

- Draw the serializability (precedence) graphs for S1 and S2
- State whether each schedule is serializable or not.
- If a schedule is serializable, write down the equivalent serial schedule(s).

T1: r1(a); r1(b); w1(a);

T2: r2(b); r2(c); w2(b);

T3: r3(c); r3(d); w3(c);

T4: r4(d); r4(a); w4(d);

S1: r1(a), r2(b), r3(c), r4(d), r1(b), r2(c), r3(d), r4(a), w2(b), w3(c), w4(d), w1(a)

S2: r4(d), r4(a), w4(d), r3(d), r3(c), w3(c), r2(b), r2(c), w2(b), r1(a), r1(b), w1(a)

Question #4:

Consider an Online Shopping database table Cart with attributes:

Assume the Total_Amount for User_ID = 101 is initially Rs 2000.

Two concurrent transactions occur:

- T1 = User applies a discount coupon of Rs 300
- T2 = User adds a new product costing Rs 800

The execution interleaves as follows:

- T1 reads Total_Amount = 2000
- T2 reads Total_Amount = 2000
- T1 subtracts 300 → computes new amount = 1700
- T2 adds 800 → computes new amount = 2800
- T1 writes 1700 to database
- T2 writes 2800 to database

Task:

- Identify the concurrency problem(s) present in this schedule. Explain how they arise.
- What is the final Total_Amount after both transactions finish?
- What would the correct Total_Amount be if the transactions were executed serially (T1 → T2 and T2 → T1)?

Question #5:

Which of the following schedules is (conflict) serializable?

For each serializable schedule, determine the equivalent serial schedules.

(a) $r_1(X); w_1(X); r_3(X); r_2(X); w_3(X);$

(b) $r_1(X); r_3(X); w_3(X); w_1(X); r_2(X);$

(c) $r_3(X); r_2(X); w_3(X); r_1(X); w_1(X);$

(d) $r_3(X); r_2(X); r_1(X); w_3(X); w_1(X);$